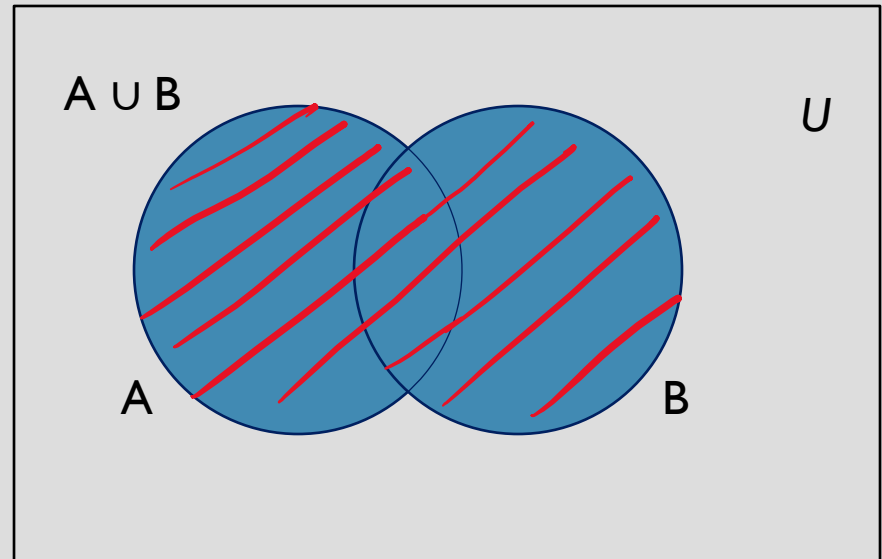


SETS

CSE 2213 – Discrete Mathematics

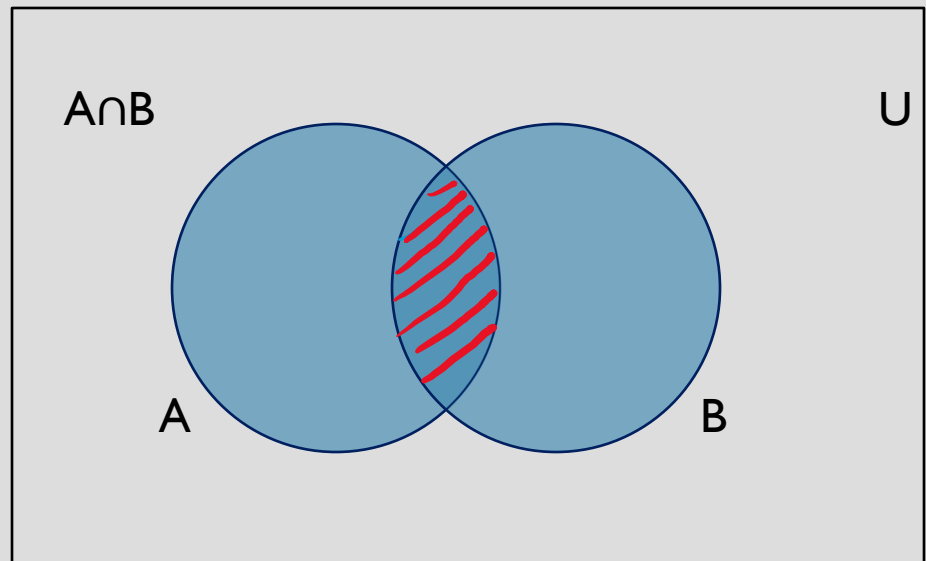
SET OPERATIONS: UNION

- $A = \{1,2,3\}, B = \{1,3,5\}$
 $A \cup B = \{1,2,3,5\}$
- $A \cup B = \{x \mid x \in A \vee x \in B\}$
- A member should exist in either of the sets or both of the sets



SET OPERATIONS: INTERSECTION

- $A = \{1,2,3\}, B = \{1,3,5\}$
 $A \cap B = \{1,3\}$
- $A \cap B = \{x \mid x \in A \wedge x \in B\}$
- A member should exist in both of the sets
- Two sets are **disjoint** if $A \cap B = \emptyset$

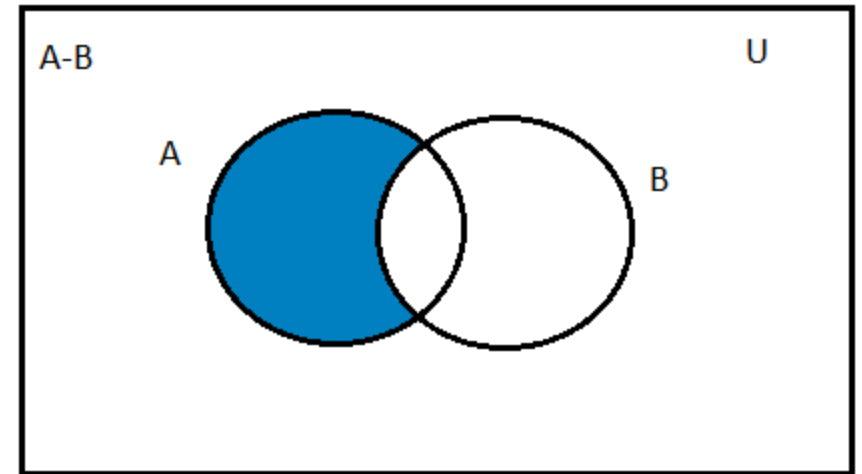


CARDINALITY OF UNION SET

- $|A \cup B| = |A| + |B| - |A \cap B|$
- Principle of inclusion and exclusion

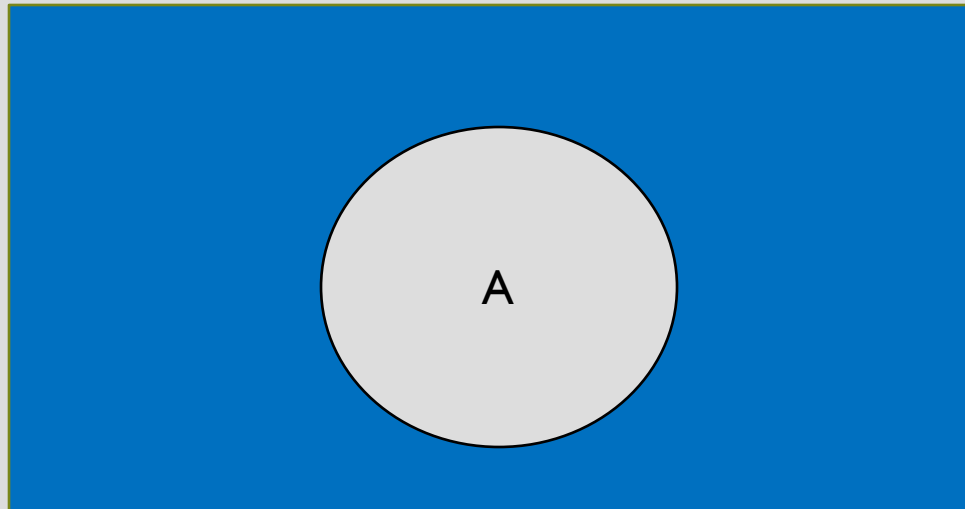
SET OPERATIONS: DIFFERENCE

- $A = \{1,2,3\}, B = \{1,3,5\}$
 $A - B = \{2\}$
- $A - B = \{x \mid x \in A \wedge x \notin B\}$
- A member should exist only in the first set



COMPLEMENT

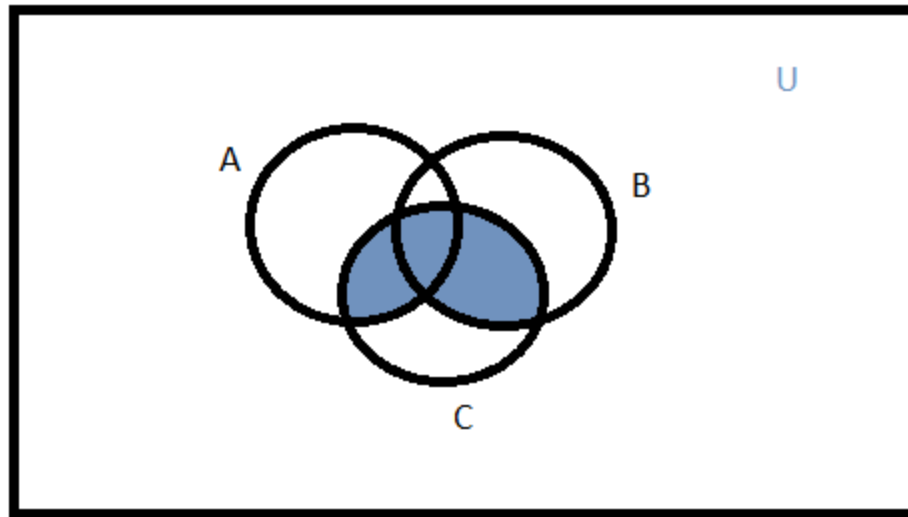
- Let U be the universal set. The complement of the set A , denoted by $\bar{A}$,
- In other words, the complement of the set A is $U - A$



PROBLEM

- Given that $A = \{0,2,4,6,8\}$, $B = \{0,1,2,3,4\}$ and $C = \{0,3,6,9\}$, find out the following:
 - $(A \cup B) - C$
 - $(A - B) \cap C$
 - Draw the Venn diagram for the following sets:
 - $(A \cup B) \cap C$
 - $(A - C) \cup B$

VENN DIAGRAM



VENN DIAGRAM

